Region-Based Relevance Feedback In Image Retrieval

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Abstract

Relevance feedback and region-based representation of images are two effective ways to improve accuracy in content-based image retrieval. In this paper, we propose a novel relevance feedback approach based on region representation. It can be considered as a special case of the query point movement method in region-based image retrieval. By assembling all the segmented regions of positive examples together and resizing the regions to emphasize the latest positive samples, we form a composite image as the optimal query. A region-based image similarity measure is used to calculate the similarity between the optimal query and an image in the database. An incremental clustering technique is also considered to improve the retrieval efficiency. Experimental results show that the proposed approach is effective in improving the performance of content-based image retrieval systems.

Keywords

Image retrieval, region-based, relevance feedback, incremental clustering, and positive example

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Relevance feedback and region-based representation of images are two effective ways to improve accuracy in content-based image retrieval. In this paper, we propose a novel relevance feedback approach based on region representation. It can be considered as a special case of the query point movement method in region-based image retrieval. By assembling all the segmented regions of positive examples together and resizing the regions to emphasize the latest positive samples, we form a composite image as the optimal query. A region-based image similarity measure is used to calculate the similarity between the optimal query and an image in the database. An incremental clustering technique is also considered to improve the retrieval efficiency. Experimental results show that the proposed approach is effective in improving the performance of content-based image retrieval systems.

1. Introduction

It is well known that the performance of content-based image retrieval (CBIR) systems is mainly limited by the semantic gap between low-level features and high-level concepts. In order to reduce this gap, two approaches have been widely used: relevance feedback to learn user's intentions and region-based features to represent the focus of user's perceptions of image content.

Most of the published works on relevance feedback [2, 4, 8, 9,11] use one or both of the following ideas: query point movement and feature re-weighting. As a representative and effective method, the MindReader [4] system has both features. The query point movement method essentially attempts to improve the retrieval results by moving the query towards positive examples and away from negative examples in the feature space.

Contrasting to traditional approaches, which compute only global features of images, the region-based methods extract features of the segmented regions and perform similarity comparisons at the granularity of region. For example, SIMPLIcity [12] uses integrated region matching (IRM) [5] as its image similarity measure. By allowing many-to-many relationship of the regions to be valid and selecting the area percentage of a region as its importance measure, IRM is robust against inaccurate segmentation.

However, little attention has been paid to combine relevance feedback and region-based methods together. Minka and Picard made a pioneer work in this area by using a "society of models" [7]. In that method, many plausible groupings of the data are first computed using a number of different filtering techniques. Based on the positive and negative examples from the user, the system modifies the relative weights of the groupings. In the IDQS by Wool et al [13], a query is initiated by the selection of a region of interest from a key image. After that, user's feedback is given in the form of acceptance or rejection of the retrieved images. Then the Learning Vector Quantization (LVQ) algorithm is employed to cluster the selected regions in feedback examples. Images with regions close to the positive cluster centroids are then returned and reclassified by the user. This iterative refinement continues until the user is satisfied with the results. In fact, it is a region-based retrieval method using classification as its feedback scheme.

Enlightened by those region-based techniques, we propose an intuitive feedback method based on region representation in this paper. The basic idea is similar to the query-point movement. By assembling all the segmented regions of positive examples together and putting more emphasis on the latest positive samples, we form an optimal query. Since it is a composite image with small regions, any region-based similarity measure can be adopted to calculate the similarity between the optimal query and an image. In order to improve the

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retrieval speed, an incremental clustering technique is also considered to merge similar regions into larger ones.

The remainder of the paper is organized as follows. In Section 2, we summarize the previous works related to region-based retrieval. In Section 3, the region-based feedback method is explained. Preliminary experimental results are given in Section 4. Finally, we conclude in Section 5.

2. Related Works

In our region-based relevance feedback approach, we employ a few related techniques, such as image segmentation, region representation, and similarity measurement. Here we briefly summarize them without going into details.

The segmentation method we use is the so-called JSEG algorithm [3]. Considering that our purpose is retrieval and the similarity measure that we use is insensitive to segmentation results, we allow the disconnected regions in comparison with the neighboring regions to be merged. In this way, we preserve the natural clustering of objects and allow compact image characterization to be defined.

We use two properties to describe a region: the feature and the importance of the region. For the former, we use color moment, which is shown to be robust and effective [10]. We extract the first three moments from each channel of HSV color space. For the latter, the percentage of region area is used, which is based on the assumption that important objects in an image tend to occupy larger areas [5].

The similarity between region R_i and R_j is defined as:

$$s(R_i, R_j) = \exp(-d(R_i, R_j)/\sigma), \quad (1)$$

where $d()$ is the Euclidean distance between the feature vectors of two regions, σ is the standard deviation of the distances computed over a sample of DB regions (about 50000 region pairs).

Based on the region similarity, the similarity between two images is measured in a way similar to IRM [5], except two modifications. Region similarity is adopted instead of distance, and a predefined threshold ε is introduced for the matching of two regions to be valid, although all region pairs are considered in the similarity measure. As a result, if two images have no valid matching, their similarity is 0. In our experiments, ε is set to 0.78. The similarity measure method is also insensitive inaccurate segmentation.

3. Region-Based Relevance Feedback

Inspired by the query-point movement method, we propose an intuitive relevance feedback approach to region-based image retrieval. The basic assumption is that every region might be helpful in retrieval and important regions should appear in more than one positive example. Based on this consideration, we may simply assemble all the regions of the initial query and the positive image examples into a single image, which is used as the optimal query in the next iteration of retrieval and feedback process. Note that negative examples are not taken into account in our approach. The optimal query is just like an over-segmented image with many small regions. After region importance normalization, important regions are enhanced, while unimportant regions are weakened in the optimal query. Since it contains all the information from the user, better retrieval results may be expected.

The above approach is simple and straightforward. It can be incorporated into any region-based image retrieval system where region-based similarity measure between images is defined. In practice, however, we have to deal with some issues related to performance, which are explained in the following.

3.1. Region Weighting

As stated above, the importance of a region is defined as the area percentage in the image. In the optimal query, the importance of each region has to be normalized such that the sum of all region importance is equal to 1. For the sake of simplicity, the initial query is treated as a positive example. There is a number of different ways to weight the region importance in order to meet the constraint. Before going into details, let us define some notations at first.

Assume that there are *n* positive examples $I_1, \cdots, I_n$, with $I_1, \cdots, I_m$ being the prior ones and $I_{m+1}, \cdots, I_n$ being the new ones. Each positive example I_k is represented by a set of regions $\{R_{k,1}, \cdots, R_{k,n_k}\}$. Let the optimal image be I^+ that contains regions $\{R_{k,i} \mid 1 \le k \le n, 1 \le k \le n_k\}$. Denote the region importance of region $R_{k,i}$ in I_k and I^+ as $RI_{k,i}$ and $RI^+_{k,i}$.

Since $\sum_{k=1}^{n}\sum_{i=1}^{n_k} RI_{k,i} = n$, to satisfy the constraint $\sum_{k=1}^{n}\sum_{i=1}^{n_k} RI^+_{k,i} = 1$, a simple way is to set $RI^+_{k,i} = RI_{k,i}/n$, which implies that all positive examples are treated equally.

At each interaction, the newly added positive examples might have more potential in finding the user-intended but undetected images, because they reflect the user's query concept more precisely. Therefore, they should have more contributions to the optimal query. A

similar idea can be found in [11], which introduces a decaying factor to reduce the effect of previous positive examples.

To reflect this bias, we can "enlarge" the newly added positive regions or equally "reduce" the prior positive regions by assigning them larger or smaller importance correspondingly.

Let $0 \le \alpha \le 1$ be a discount factor that attenuates the importance of the prior positive examples. The larger α is, the lower the importance of the prior positive regions will be. The total region importance of all the positive examples is:

$$(1-\alpha)\sum_{k=1}^{m}\sum_{i=1}^{n_k} RI_{k,i} + \sum_{k=m+1}^{n}\sum_{i=1}^{n_k} RI_{k,i}$$
$$= (1-\alpha)m + n - m = n - \alpha \cdot m$$

Considering the constraint $\sum_{k=1}^{n}\sum_{i=1}^{n_k} RI^{+}_{k,i} = 1$, we set the region importance of the optimal query I^{+} as follows:

$$RI^{+}_{k,i} = \frac{RI_{k,i} * (1-\alpha)}{n - \alpha * m}, \text{ if } 1 \le k \le m,\ 1 \le i \le n_k,$$

$$RI^{+}_{k,i} = \frac{RI_{k,i}}{n - \alpha * m}, \text{ if } m+1 \le k \le n,\ 1 \le i \le n_k.$$

Note that if $\alpha = 0$, $RI^{+}_{k,i}$ will be equal to $RI_{k,i}/n$, which is the case discussed before. For another extreme, $\alpha = 1$, all the prior positive regions are completely ignored. If $\alpha = 1 - 1/m$, all the prior positive examples are deemed as just one image:

$$(1-\alpha)\sum_{k=1}^{m}\sum_{i=1}^{n_k} RI_{k,i} = (1-\alpha)m = 1.$$

3.2. Region Clustering

As more positive examples are available, the number of regions in the optimal query increases rapidly. As the time required calculating the image similarity is proportional to the number of regions in the query, the retrieval speed of the system will slow down gradually. To avoid this, we have proposed a practical and effective way to cut down the region number of the optimal query by using some clustering method. This process is similar to the region merging in an over-segmented image.

Because of its simplicity, the k-means algorithm is used to cluster the regions of all positive examples into a number of classes, each of which is corresponding to a new region of the optimal query. We adaptively choose the number of clusters k by gradually increasing its value. Since the method used to calculate the image similarity is robust against inaccurate segmentation, the value of k is not very critical. Therefore, a small k is preferred. k is set to 2 at the start and increases by one at each step. This process stops when the similarity between any positive region and its nearest cluster is above a predefined threshold ε, which is introduced in Section 2.

Based on the consideration that all prior positive examples are unchanged as new ones are added, we can utilize the clustering information in previous iteration to further accelerate the current clustering process. In particular, we use the centroids of previous clusters, which are also the regions of the prior optimal query, to represent all the prior positive examples. Note that the computational cost of k-means is $O(kn)$ for the data size n, and the number of clustering centroids is much smaller than the number of positive regions. In this way, the computation time in clustering is significantly reduces. This method is referred to as incremental region clustering.

After clustering, regions in a cluster are merged into a larger one, and their normalized region importance values are summed together to be the region importance of the merged region. Meanwhile, the feature representation of this new region has to be formed. A general way is to extract the features on the merged region directly. However, the feature extraction is a time-consuming process, and is usually done off-line. Thus, we adopt a simple alternative instead. Considering that the regions in the same cluster are very similar, we may use the average feature of individual regions as the feature of the new region.

3.3. Relevance Feedback Method

We summarize the feedback method in our implementation as follows.

The initial query is regarded as a positive example. At the first iteration of feedback, all regions of positive examples are assembled into a composite image, in which similar regions are grouped into clusters by k-means algorithm. Regions within a cluster are merged into a new region. The feature of the new region is equal to the average feature of individual regions, while the importance of it is set to the sum of individual region importance divided by the number of positive examples. This composite image is used as the optimal query example.

In the following iterations, only the optimal query is treated as a positive example with all other prior examples being ignored. That is, there is exactly one prior positive example, which is treated equally as newly added positive examples. This implies that the importance of the prior positive examples gradually decays in the optimal query, and the importance of the newly added ones is emphasized accordingly.

4. Experimental Results

To evaluate the performance of the system, we used a general-purpose image database (from COREL) of about 8,600 images as the ground truth. In our experiments, 20 images from each of ten selected groups of images were randomly chosen as query images, which resulted in 200 trials. The selected groups are: beach, card, eagle, firework, flower, fashion model, ski, plane, mountain and subset. For each of these 200 query images, 10 iterations of user-and-system interaction were carried out. At each iteration, the system examined the 50 most similar images to the optimal query, except those positive examples labeled in previous iterations. Images from the same category as the initial query image were used as new positive examples. At next iteration, all positive images were placed in top ranks directly, while others were compared with the optimal query again and ranked according to their similarities.

Considering that it is difficult to design a fair comparison with existing systems that use region-based relevance feedback, such as the FourEyes [7] system whose propose is annotation and the IDQS [13] system that depends on manually defined queries, we compared our method with that used in MindReader [4]. To be fair, we started each query with the same feature (color moments) and the same distance metric (Euclidean) in both methods. The average precision within the fist 50 retrieved images (P (50)) is plotted against the number of iterations. The result is shown in Figure 1. As shown in the figure, our method consistently yields better performance after one interaction and its average P (50) after 9 interactions is higher than that of MindReader by 26%.

To show the effectiveness of our emphasis on latest positive examples, we carried out the experiment under two situations: with bias and no bias. The results are shown in Figure 2. It is easily seen from the result that by making the new positive examples more important, the retrieval accuracy is improved substantially.

The algorithm has been implemented on a Pentuim III 550 PC using the Win2000 operating system. The average retrieval time and average p (50) of our method with no clustering and incremental clustering are compared and shown in Figure 3 and Figure 2 correspondingly. As we can see from both figures, the proposed clustering scheme can save much retrieval time without decreasing the retrieval accuracy.

5. Conclusion

In this paper, we have presented a novel relevance feedback method based on region representation. The basic idea is to properly assemble all the positive examples into a segmented image referred as the optimal query. In order to emphasize the latest positive examples that express the user's query concept more precisely, we have designed an effective weighting scheme. Considering that the optimal query is over-segmented, which may cost too much time to calculate the similarity, we have introduced an incremental clustering method that increases the retrieval speed significantly. Experimental results on a database of about 8,600 general-purpose images show the effectiveness of the emphasis on new positive examples and the superiority of our method over others.

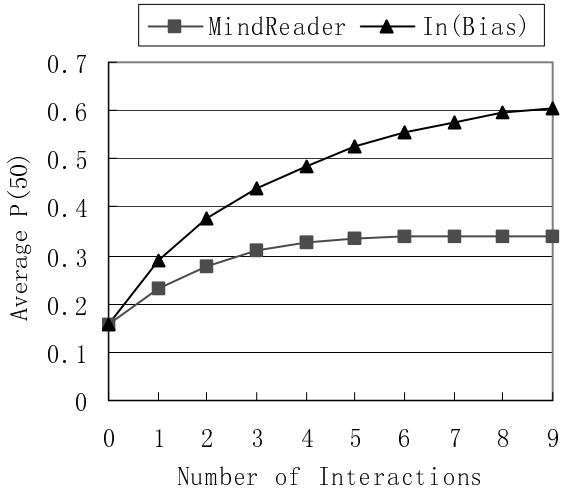


Figure 1. **Retrieval accuracy comparison of MindReader and our method using Incremental clustering and bias on new positive examples.**

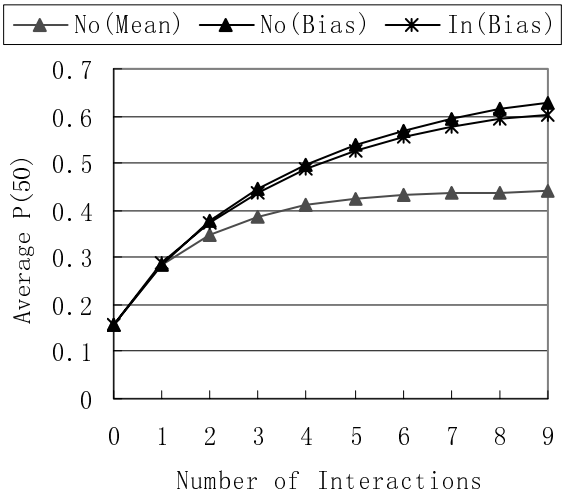


Figure 2. **Performance Comparison of using bias (bias) and not using bias (mean) under two situations: no clustering (no) and incremental clustering (in).**

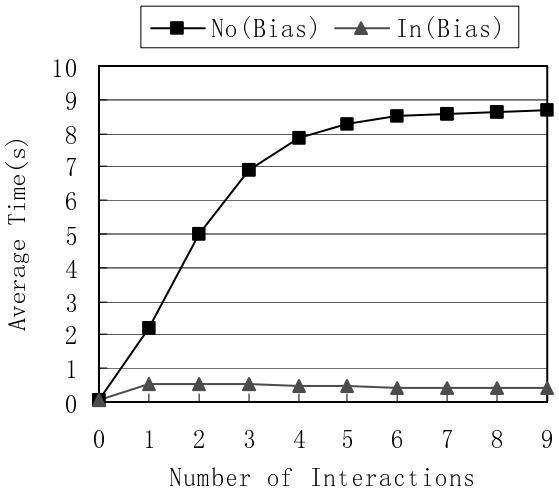


Figure 3. **Average retrieval time of our method using bias under two situations: no clustering (no) and incremental clustering (in).**

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